Aashray Jain

Software Engineer | Full-Stack & AI/ML Specialist

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Summary

Full-Stack & AI/ML Engineer with 4+ years of professional experience across enterprise cloud backends and production LLM orchestration. Key contributor at Novafy OU rebuilding NovaAgent v2 in Python/FastAPI with Model Context Protocol (MCP) tool interfaces, prompt-caching (∼99% cached tokens), and ERP Human-in-the-Loop validation. Former TCS Systems Engineer delivering BFSI real-time risk platforms (500K+ users) with 70% cloud cost reduction. Inventor of ArihantQNet (tri-branch XAI architecture) and registered copyright holder for ArihantQInvoice ERP. Pursuing M.Tech. in AI/ML at NIMS University.

Education

- **M.Tech. in Artificial Intelligence & Machine Learning** *2026 – 2028*
  NIMS Marik Institute of Computing, AI, Robotics & Cybernetics, NIMS University Rajasthan Pursuing
- **B.Tech. in Electronics & Communication Engineering (8.74 CGPA)** *2017 – 2021*
  Lakshmi Narain College of Technology (LNCT), Bhopal (RGPV Bhopal) Katni, MP

Experience

- **Novafy OU** *Oct 2026 – Present*
  Software Development Intern Remote (Hamburg, Germany)
  - Core contributor rebuilding the **NovaAgent v2 conversational engine** in Python, FastAPI, and PostgreSQL (190+ commits) with per-tenant configurations, dynamic multi-model fallback chains, and database-backed model catalogs.
  - Architected tool interfaces consuming and serving capabilities over **Model Context Protocol (MCP)**; built a **Human-in-the-Loop (HITL) safety layer** for Dynamics 365 Business Central ERP with live metadata validation and code substitution.
  - Optimized latency and operational costs via provider-side **prompt-caching achieving ∼99% cached input tokens**; developed conversation compaction and fit-to-window algorithms preventing context overflow.
  - Built automated evaluation harness with 70+ conversational test scenarios to benchmark prompts and models; hardened tenant isolation, replay protection, and idempotent MS Teams message handling.
- **ArihantQ** *Aug 2025 – Present*
  Founder (Part-Time and Experimental Venture) Remote (India)
  - Architected ArihantQInvoice, a production full-stack ERP for billing, inventory, and supplier workflows using Node.js, Express, SQLite, and Sequelize; officially registered software copyright with the Copyright Office, Govt. of India.
  - Validated technical concepts in national hackathons: prize winner in Lyzr AI Architect Challenge (AI Cost Optimization Advisor); public recognition from MariaDB CEO and Foundation; secured legal business enablers (Udyam, Shop & Establishment License).
- **Tata Consultancy Services** *Jul 2021 – Jan 2025*
  Systems Engineer Mumbai & Kochi, India
  - Built secure real-time alert analysis systems and LOB dashboards for a Canadian BFSI client using React, Node.js Lambda, and DynamoDB, serving 500K+ enterprise users and boosting risk monitoring efficiency by 90%.
  - Led a multi-account AWS cloud migration reducing infrastructure costs by 70%; automated reporting workflows using Angular, Python (Pandas/Flask), and Excel automation, saving 15+ hours/week; collaborated with TCS Rapid Labs on enterprise AI/ML PoCs.

Projects

- **Class-Balanced Benchmarking & XAI Clinical DSS (ArihantQNet)** *Oct 2026*
  PyTorch, Python, Streamlit, Grad-CAM, Scikit-learn Live App | GitHub
  - Stratified multi-stage Alzheimer's Disease on 6,400 MRI scans; engineered Normalized Square-Root Inverse-Frequency Weighted Loss, achieving 100% sensitivity on rare Moderate Demented class.
  - Formulated novel tri-branch hybrid **ArihantQNet** fusing ResNet50, DenseNet121, and EfficientNet-B0 with Quantum-Inspired Cross-Attention (Q-MCA); integrated Grad-CAM neuroanatomical heatmaps in an interactive Streamlit workbench.
- **ArihantQInvoice Billing & Inventory Management System** *Apr 2026*
  Node.js, Express.js, SQLite, Sequelize, EJS, Swagger
  - Deployed production ERP platform for retail operations; built 50+ REST APIs covering GST billing, dynamic pricing, and automated DOCX generation; officially registered software copyright with Copyright Office, Government of India.

Technical Skills

- **Languages & Backend:** Python, JavaScript (ES6+), TypeScript, SQL, Node.js, Express.js, FastAPI, REST APIs, Microservices, PostgreSQL, SQLite
- **AI / ML & GenAI:** Model Context Protocol (MCP), PyTorch, LangChain, Hugging Face, FAISS, Prompt-Caching, Streamlit, Grad-CAM (XAI), HITL
- **Cloud, Security & DevOps:** AWS (Lambda, API Gateway, S3, Cognito, DynamoDB, CDK), GCP (Vertex AI), Docker, Git, CI/CD, SAML, RBAC
- **Ecosystems & Frontend:** Microsoft 365, MS Teams App, Dynamics 365 Business Central, SharePoint | ReactJS, Angular, HTML5, CSS3, D3.js

Honors & Key Certifications

- **Honors:** Top 100 Winner Microsoft AI Odyssey Challenge (2024); 1920-min Azure AI Solution (AI-102T00); Lyzr AI Challenge Winner; MariaDB Recognition.
- **Certifications:** **Microsoft**: GenAI with Azure OpenAI | **AWS**: Cloud Practitioner Essentials | **Google Cloud**: GenAI Apps with Gemini & Streamlit | **IBM**: Applied Data Science with Python | **NPTEL**: IIT Kharagpur Java | **SEBI**: Certified